

Overblik over Hesel ligningerne

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1 Standard form

HESEL ligningerne er givet på følgende form:

$$d_t n + n\mathcal{K}(\phi) - \mathcal{K}(p_e) = \Lambda_n \quad (1)$$

$$d_t^0 \omega^* + \{\phi, p_i\} - \mathcal{K}(p_e + p_i) = \Lambda_{\omega^*} \quad (2)$$

$$\frac{3}{2}d_t p_e + \frac{5}{2}p_e \mathcal{K}(\phi) - \frac{5}{2}\mathcal{K}\left(\frac{p_e^2}{n}\right) = \Lambda_{p_e} \quad (3)$$

$$\frac{3}{2}d_t p_i + \frac{5}{2}p_i \mathcal{K}(\phi) + \frac{5}{2}\mathcal{K}\left(\frac{p_i^2}{n}\right) - p_i \mathcal{K}(p_e + p_i) = \Lambda_{p_i} \quad (4)$$

2 Lambda termer

Lambda termerne er givet ved:

$$\begin{aligned} \Lambda_n = & D_e \nabla_{\perp}^2 n - \frac{n}{\tau} - \frac{n - n_p}{\tau_p} \\ & - \alpha \left(\tilde{T}_e + \frac{\bar{T}_e}{\bar{n}} \tilde{n} - \tilde{\phi} \right) \end{aligned} \quad (5)$$

$$\begin{aligned} \Lambda_{\omega^*} = & D_i \nabla_{\perp}^2 \omega^* - \frac{\omega^*}{\tau} + \frac{\rho_s}{L_c} \left[1 - \exp \left(\phi_m - \frac{\phi_s}{T_{e,s}} \right) \right] \\ & - \alpha \left(\tilde{T}_e + \frac{\bar{T}_e}{\bar{n}} \tilde{n} - \tilde{\phi} \right) \end{aligned} \quad (6)$$

$$\begin{aligned} \Lambda_{p_e} = & \frac{5}{2} D_e \nabla_{\perp}^2 p_e + \left(\frac{16}{6} - \frac{5}{2} \right) \nabla \cdot (n \nabla_{\perp} T_e) - \frac{9p_e}{2\tau} - \frac{T_e}{\tau^{SH}} - \Theta - \frac{p_e - p_{e,p}}{\tau_p} \\ & - \bar{T}_e \alpha \left(\tilde{T}_e + \frac{\bar{T}_e}{\bar{n}} \tilde{n} - \tilde{\phi} \right) \end{aligned} \quad (7)$$

$$\Lambda_{p_i} = D_i \nabla_{\perp}^2 p_i - D_i T_i \nabla_{\perp}^2 n - \frac{9p_i}{2\tau} + \Theta - \frac{p_i - p_{i,p}}{\tau_p} + p_i \Lambda_{\omega^*} \quad (8)$$

3 Funktioner

Funktionerne er givet ved:

$n : n(x, y, t) :$	Tæthed
$p_{e,i} : p_{e,i}(x, y, t) :$	Elektron- og iontryk
$T_{e,i} : T_{e,i}(x, y, t) = p_{e,i}/n :$	Elektron- og iontemperatur
$\phi : \phi(x, y, t) :$	Elektrisk potentiale
$\omega^* : \omega^*(x, y, t) = \nabla^2 \phi + \nabla^2 p_i :$	Vorticitet

4 Operatorer

Herved er operatorerne defineret med:

$$\mathcal{K}(f) = -\frac{\rho_s}{R}\partial_y f \quad (9)$$

$$\{f, g\} = \partial_x f \partial_y g - \partial_y f \partial_x g \quad (10)$$

$$\nabla = (\partial_x, \partial_y, \partial_z)^T \quad (11)$$

$$\nabla_\perp = (\partial_x, \partial_y, 0)^T \quad (12)$$

$$\nabla^2 = (\partial_x^2, \partial_y^2, \partial_z^2)^T \quad (13)$$

$$\nabla_\perp^2 = (\partial_x^2, \partial_y^2, 0)^T \quad (14)$$

$$\mathrm{d}_t^0 f = \partial_t f + \{\phi, f\} \quad (15)$$

$$\mathrm{d}_t f = \partial_t f + \frac{1}{B} \{\phi, f\} \quad (16)$$

$$\bar{f} = \frac{1}{L_y} \int_0^{L_y} f \mathrm{d}y \quad (17)$$

$$\tilde{f} = f - \bar{f} \quad (18)$$

$$(19)$$

5 Koeffecienter

Samling af koeffecienter:

$$\Theta = \frac{3m_e}{m_i} v_{ei} (p_e - p_i) \quad (20)$$

$$\alpha = \frac{2T_{e0}}{v_{ei} m_e L_\parallel^2 \omega_{i0}} \quad (21)$$

$$\phi_m = \ln(\sqrt{m_i/2\pi m_e}) \quad (22)$$

$$(23)$$

6 Profile region

Hvor er profile regionen defineret ved:

$$\frac{n - n_p}{\tau_p} \sigma_{lcf s} \quad (24)$$

$$\frac{p_e - p_{e,p}}{\tau_p} \sigma_{lcf s} \quad (25)$$

$$\frac{p_i - p_{i,p}}{\tau_p} \sigma_{lcf s} \quad (26)$$

$$\frac{\phi_s}{T_{e,s}} = \frac{\phi(x, y, t)}{T_e(x, y, t)} * \sigma_{connected} \quad (27)$$

$$\frac{\phi_s}{T_{e,s}} = \frac{\bar{\phi}(x, y, t)}{\bar{T}_e(x, y, t)} * \sigma_{disconnected} \quad (28)$$

$$(29)$$

$$\sigma_{lcf s} = \begin{cases} 1 & \text{for } x < x_{lcf s} \\ 0 & \text{for } x > x_{lcf s} \end{cases} \quad (30)$$

$$\sigma_{connected} = \begin{cases} 1 & \text{for } x < x_{connected} \\ 0 & \text{for } x \geq x_{connected} \end{cases} \quad (31)$$

$$\sigma_{disconnected} = \begin{cases} 1 & \text{for } x < x_{disconnected} \\ 0 & \text{for } x \geq x_{disconnected} \end{cases} \quad (32)$$

7 parametre

Parametre for HESEL ligningerne:

Log fil def	Symbol	Værdi	Bruges
*bt	B	1.11	9, 16
me###	m_e	$9.109382e - 31$	20, 21, 22
mi#	m_i	$3.345243e - 27$	20, 22
Rmajor##	R	$8.800000e - 01$	9
rhos	ρ_s	$8.929325e - 04$	9, 6
Te0##	T_{e0}	$2.980000e + 01$	21
nuei#x8	v_{ei}	$4.379143e + 06$	20, 21
De#x8	D_e	$6.654814e - 02$	5, 7
#Di##	D_i	$2.851611e + 00$	6, 8
*floor_n	n_p	0.005	5
*floor_pe	$p_{e,p}$	0.000025	7
*floor_pi	$p_{i,p}$	0.000025	8
*floor_time	τ_p	50	5, 7, 8
*force_time	τ	50	5, 6, 7, 8
*x_lcf	σ_{lcf}	0.4	30
*x_connected	$\sigma_{connected}$	0.4	31
*x_disconnected	$\sigma_{disconnected}$	0.4	32

Alt er taget fra BOUT.log.0, og er derfor fundet ved at erstatte det der står i log filen med det der står i tabellen. Hvis der står * skal parametret prefixes med <Option hesel:>

Hvis der står # findes parametret ved at erstatte # med mellem rum